

NETWORK MONITORING SYSTEM (NMS)

Enterprise-Grade, Real-Time Network Visibility & Control

1. PRODUCT OVERVIEW

The **Network Monitoring System (NMS)** is a comprehensive, scalable, and intelligent platform designed to provide **real-time visibility, performance monitoring, fault detection, and automated alerting** across your entire network infrastructure — from core routers and switches to endpoints, servers, cloud instances, and IoT devices.

2. KEY FEATURES & FUNCTIONS

A. Network Discovery & Mapping

Feature	Description
Auto-Discovery	Automatically discovers all network devices via SNMP (v1/v2c/v3), ICMP, WMI, SSH, and NetFlow, sFlow, jFlow (via Probe, Network Traffic Analysis Module)
Topology Mapping	Dynamic Layer 2/Layer 3 topology maps with drag-and-drop customization
Device Inventory	Centralized inventory with device type, OS, firmware, serial number, and location
Subnet Scanning	Scheduled or on-demand scanning of IP ranges and subnets
Dependency Mapping	Visual parent-child relationship mapping to reduce false alerts

A. Real-Time Performance Monitoring

Feature	Description
Bandwidth Monitoring	Real-time bandwidth usage tracking per interface (In/Out/Total)
Latency & Jitter	End-to-end latency, jitter, and packet loss measurement
CPU / Memory / Disk	Resource utilization monitoring for servers, routers, switches, and firewalls
Interface Monitoring	Port-level status, errors, discards, speed, and duplex monitoring
WAN / MPLS Monitoring	Circuit monitoring with SLA threshold tracking
NetFlow / sFlow / jFlow	Deep traffic analysis — Top Talkers, Top Applications, Top Protocols (via Probe, Network Traffic Analysis Module)
QoS Monitoring	Monitor DSCP markings and traffic class performance

B. Alerting & Notification Engine

Feature	Description
Threshold-Based Alerts	Multi-level thresholds (Warning / Critical / Fatal)
Intelligent Alerting	Alert correlation, de-duplication, and root-cause analysis
Escalation Policies	Multi-tier escalation with time-based rules

Feature	Description
Notification Channels	Email, SMS, Webhook, SNMP Traps
Alert Suppression	Maintenance windows and dependency-based suppression
Custom Alert Rules	Compound conditions using AND/OR logic across multiple metrics

C. Reporting & Analytics

Feature	Description
Pre-Built Reports	100+ out-of-the-box reports (Availability, Health, Traffic, Top-N)
Custom Report Builder	Drag-and-drop report designer with scheduling and export (PDF/CSV/Excel)
SLA Reports	Uptime/downtime SLA compliance reports with breach tracking
Capacity Planning	Historical trend analysis with pattern analysis charts and graphs
Executive Dashboards	Role-based dashboards with real-time KPIs and widgets

D. Configuration & Security Management (Optional if Selected)

Feature	Description
Config Backup	Automated scheduled backup of running/startup configurations
Config Change Detection	Real-time alerts on unauthorized configuration changes
Config Comparison	Side-by-side diff comparison of configuration versions

Feature	Description
Compliance Auditing	Policy-based compliance checks (PCI-DSS, HIPAA, SOX baselines)
Firmware Management	Centralized firmware version tracking and upgrade scheduling

E. Multi-Environment Support

Feature	Description
Cloud Monitoring	AWS, Azure, GCP — EC2, VPC, Load Balancers, Cloud Firewalls
Virtual Infrastructure	VMware vSphere, Microsoft Hyper-V, KVM monitoring
SD-WAN Monitoring	Support for Cisco Viptela, Fortinet, Meraki, VeloCloud
Wireless Monitoring	Access Points, Controllers, SSID performance, client count
IoT Device Monitoring	Support for custom SNMP MIBs and API-based polling

F. Integration & Extensibility

Feature	Description
RESTful API	Full-featured REST API for CRUD operations, data queries, and automation
ITSM Integration	Any ITIL v3 and ITIL v4 Product — auto-ticket creation on alerts
SIEM Integration	Syslog forwarding
Automation	Ansible, Python scripting, and built-in workflow automation

Feature	Description
SSO & LDAP	Active Directory, LDAP, SAML 2.0, OAuth 2.0 authentication

3. TECHNICAL SPECIFICATIONS

Parameter	Specification
Supported Devices	50 to 100,000+ nodes (tiered licensing)
Polling Interval	Configurable: 30 sec to 24 hrs (default: 5 min)
Protocols Supported	SNMP v1/v2c/v3, ICMP, WMI, SSH, Telnet, HTTP/HTTPS, NetFlow v5/v9, IPFIX, sFlow, jFlow
Database	PostgreSQL / MySQL / MS SQL (varies by deployment)
High Availability	Active-Passive Failover, Distributed Polling Architecture
Data Retention	Configurable: 30 days to 5+ years (with data roll-up)
Web Console	HTML5-based, responsive — Chrome, Firefox, Edge supported

4. SYSTEM REQUIREMENTS (MINIMUM)

Component	Small (≤500 devices)	Medium (≤5,000 devices)	Large (≤50,000+ devices)
CPU	4 Cores	8 Cores	16+ Cores
RAM	8 GB	32 GB	64+ GB

Component	Small (≤500 devices)	Medium (≤5,000 devices)	Large (≤50,000+ devices)
Storage	100 GB SSD	500 GB SSD	2+ TB SSD/NVMe
Network	1 Gbps	1 Gbps	10 Gbps
Architecture	Single Server	Primary + Remote Poller	Distributed + HA Cluster

5. SUPPORT & SERVICES

Service	Details
Technical Support	Business Hours (Professional)
Updates	Quarterly feature releases + monthly security patches
Onboarding	Professional deployment & migration services available
Training	Online courses, certifications, and knowledge base

6. COMPETITIVE HIGHLIGHTS

	Advantage
	Unified Monitoring — Network, Server, Cloud, Wireless in one pane
	Fast Deployment — Operational within hours, not days
	AI-Powered Insights — Anomaly detection & predictive analytics

	Advantage
	Scalable — From SMB (50 nodes) to Enterprise (100K+ nodes)
	Cost Effective — Lower TCO vs. fragmented point solutions
	Vendor Agnostic — Supports 10,000+ device types out-of-the-box